

Old fashion P.T.: $H = H_0 + V$

$$H_0 |\phi\rangle = E |\phi\rangle$$

• Can find eigenstates $|\psi\rangle$ s.t.

$$H |\psi\rangle = E |\psi\rangle$$

formally:

$$H |\psi\rangle = (H_0 + V) |\psi\rangle = E |\psi\rangle$$

$$H_0 |\phi\rangle = E |\phi\rangle \Rightarrow (H_0 - E) |\phi\rangle = 0$$

$$(H_0 - E + V) |\psi\rangle = 0$$

add them
...

$$(H_0 - E)(|\psi\rangle - |\phi\rangle) + V |\psi\rangle = 0$$

$$\text{Formally: } |\psi\rangle = |\phi\rangle + \left(\frac{V}{H_0 - E} \right) |\psi\rangle = 0$$

$$|\psi\rangle = |\phi\rangle + \frac{V}{H_0 - E} |\psi\rangle$$

Lippmann-Schwinger Eq.

* Lippmann-Schwinger Kernel:

$$\tau_{LS} = (E - H_0)^{-1}$$

Define an operator τ by $V |\psi\rangle = \tau |\phi\rangle$

$$|\psi\rangle = |\phi\rangle + \frac{\tau |\phi\rangle}{E - H_0}$$

$$V |\psi\rangle = V |\phi\rangle + V \frac{1}{E - H_0} \tau |\phi\rangle$$

$$\tau |\phi\rangle = \left(V + V \frac{1}{E - H_0} \tau \right) |\phi\rangle$$

$$\therefore T = V + V \left(\frac{1}{E - H_0} \right) T$$

so

$$T = V + V \frac{1}{(E - H_0)} \left(V + V \frac{1}{E - H_0} T \right)$$

$$= V + V \frac{1}{E - H_0} V + V \frac{1}{E - H_0} V \frac{1}{E - H_0} T$$

$$= V + V \Pi_{LS} V + V \Pi_{LS} V \Pi_{LS} T$$

...